Design Study on a 9.2-T NbTi Superconducting Magnet With Long-Length Uniform Axial Field

Yinming Dai, Hui Wang, Xinning Hu, Shunzhong Chen, Junsheng Cheng, Chunyan Cui, Housheng Wang, Baozhi Zhao, Yi Li, Luguang Yan, and Qiuliang Wang

***Abstract*—A NbTi superconducting magnet with long-length uniform axial field is designed and will be fabricated for high-frequency gyrotron experiments. The magnet has a warm bore of 90 mm in diameter and a uniform axial field length of 200 mm. The magnetic field on the 200-mm axial length is adjustable from 0 to 9.2 T, accommodating a wide range of gyrotron experiment frequencies. The magnet is composed of ten coaxial NbTi coils connected in series. Optimization of the field configuration is carried out to obtain the magnet parameters. Cryogenics of the magnet system is adopted as conduction cooling with a single GM cryocooler. The magnet design and integration are presented in this paper.**

***Index Terms*—Superconducting magnet, GM cryocooler, NbTi, conduction-cooled, magnetic field optimization.**

I. Introduction

SUPERCONDUCTING magnet is a powerful tool to investigate fundamental phenomenon of high energy particles under specified magnetic field. The most applications are the high energy physics experiments such as accelerators [1], [2]. Other research related with magnetic field is nuclear magnetic resonance (NMR), which studies resonance of proton under very uniform magnetic field [3]. For the purpose of radiation from curved electrons, such as wiggler and gyrontrons, magnetic field with oscillating or gradient distribution are required to guide the charged particles [4]–[6].

For the gyrotron applications, the primary task of the magnet is to provide a magnetic field to guide the cyclotron electron procession along the magnetic line of force, from the cathode to the interaction region, where the magnetic field is uniform and coherent electromagnetic radiation emits at approximately the electron cyclotron frequency. The cyclotron frequency is a function of magnetic field and electron energy, depending on the specified applications. For example, a 140–170 GHz gyrotron oscillator, which is used as plasma heating in Tokamak fusion experiment, employs the magnetic field of 4.7–5.8 T as the interaction field [7]–[10]. Higher resonant frequencies of cyclotron electron need stronger magnetic field to harness the processional motion. As a promising technique, the THz radiation source can be used to detect the internal physics information of semiconductor, dielectric film and other body materials. In 2008, authors from University of Fukui [11] had reported that the highest frequency of 1.08 THz was achieved at second harmonic with an experiment apparatus involving a 20-T superconducting magnet. The application experiments to high-frequency Dynamic Nuclear Polarization (DNP) NMR were mentioned. Terahertz gyrotron with fundamental wave experiments needs much stronger magnetic field, saying 36–40 T, which require a huge investment to build such a high-field superconducting magnet. So, second or multiple harmonic gyrotrons are accepted to realize high frequency using a moderate magnetic field.

Low cost superconducting magnet with moderate magnetic induction becomes popular for the high-power, high-frequency gyrotron users, which can meet the requirements of high-frequency experiments with multiple harmonics.

The purpose of our development of a 9.2-T superconducting magnet is to meet the requirements of gyrotron users to fulfill a wide range of frequency experiments, such as fundamental gyrotron oscillators of 140 GHz (5.8 T), 250 GHz (9 T), and second harmonic 460 GHz (8.4 T). Features of the magnet include a full NbTi superconducting coil groups, adjustable axial magnetic field distribution and conduction-cooled cryogenics as well as a long uniform magnetic field length which is beneficial to perform experimental frequencies with long wavelengths. Detailed magnetic structure design and construction are described in the following paragraphs.

II. Superconducting Coils and Magnetic Fields

A. Superconducting Coils

Design of the 9.2-T NbTi superconducting magnet should meet the user requirements as follows:

warm bore diameter: 90 mm;
magnetic field in the warm bore: adjustable from 0 to 9.2 T;
magnetic field length: no shorter than 200 mm;
magnetic field at cathode: ~3000 Gs.

The maximum available magnetic field of 9.2 T in the warm bore implicates that the magnet can be accomplished with NbTi material as long as the peak magnetic field inside the coils falling in its critical surface. Peak field optimization is a crucial issue in the magnet design. Fig. 1 shows the optimized coils profiles, which produce an axial magnetic field distribution shown in Fig. 2.

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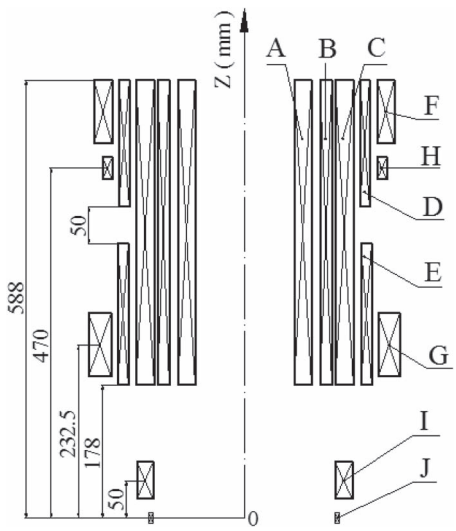

Fig. 1. Superconducting coils of the magnet.

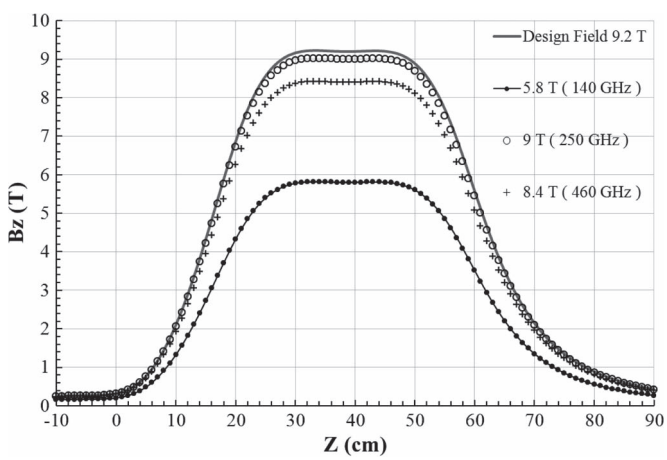

Fig. 2. Axial field distribution of design and other possible application field.

As seen from Fig. 1, the 9.2-T superconducting magnet is composed of 10 NbTi coils grouped in two parts. The first part is the main magnet producing the designated 9.2-T central magnetic induction at an operating current of 68.7 A. It consists of 8 coaxial coils (from A to G) arranged in separate layers. The second part is at the bottom of the main magnet, consisting of two small coils. It is functioned as the field adjusting coils for restraining the emitted electrons from cathode. Operating current of the adjusting coils is nearly the same as that of the main magnet. Two power supplies are necessary to charge the main magnet and adjusting coils, respectively. The uniform magnetic field length of 200 mm can be reached which corresponding to a long interaction region for gyrotron experiment. Detailed parameters of these coils are listed in Table I.

Magnetic field analysis is performed based on the designed coil parameters. The peak fields in the main magnet are 9.5 T, 8.1 T, and 6.9 T for coils A, B, and C, respectively. Hoop stresses are assessed for each of the coils by using finite element method. Maximum hoop stresses for coils A, B, and C are 62 MPa, 88 MPa, and 77 MPa, respectively. The peak hoop stress is 90 MPa which is located at the innermost windings of coil E. The reason of the peak stress in coil E can be attributed to the fact that there exists a higher axial field superposed by the outer thick coil G. Hoop stresses in the other coils are not so prominent that no special attentions should be taken.

TABLE I
COIL PARAMETERS OF THE 9.2-T MAGNET

Coil No.	Inner Dia. mm	Outer Dia. mm	Length mm	Curr. Density A/mm^2
A	140.0	185.0	410.0	62.8
B	198.0	230.0	410.0	85.5
C	230.5	277.5	410.0	108.3
D	290.5	317.9	170.5	141.4
E	290.5	317.9	189.5	141.4
F	323.9	371.3	85.0	229.0
G	323.9	382.1	85.0	229.0
H	323.9	349.8	30.0	229.0
I	323.9	349.8	30.0	229.0
J	240.0	250.8	15.0	229.0

B. Superconductor and Operation Estimation

The maximum operating magnetic field of the magnet is 9.2 T along the magnet axis. From the point of view of material utilization, gradient current densities are adopted in the design which might reduce magnet cold mass effectively. The peak magnetic induction in the coils is 9.5 T, which is approaching the upper critical field of NbTi material at liquid helium temperature. In assessing the magnet operation performance, both the critical current density and the upper critical field of NbTi should be scrutinized in choosing suitable NbTi wires. The magnet performance is dominated by the peak magnetic induction in the coils, which must have no less than 5 percent operating margins when evaluating the critical performance of NbTi superconducting wires.

The starting point of designing the 9.2-T NbTi superconducting magnet is to use conduction cooling cryogenics. Unlike the liquid helium bath cooling, which the operating temperature is heat sunk to a stable 4.2 K, the conduction cooling depends on a thermal balance between the cooling power of a GM cryocooler and the heat leakage of thermal radiation, conductions from the surrounding room temperature cryostat. In practice, the magnet operating temperature may deviate somewhat from the liquid helium temperature. It is necessary to examine the critical properties of NbTi wire before winding, especially the high-field critical current densities at possible operating temperatures.

For commercial Nb47Ti superconducting wires provided in mass production for MRI magnet applications, the high critical current density Jc is emphasized at 5 T and 4.2 K, which levels above 3000 A/mm^2 by adopting the Artificial Pinning Center (APC) technique [12], [13]. APC is very effective in enhancing Jc at low field region, but seems not positive in improving the upper critical field Hc2. The high-field Jc NbTi wire can be obtained by adopting a high niobium content composition [14], [15]. It provides an alternative to use high-field NbTi wire for the coil A where the peak field locates.

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TABLE II
SPECIFICATIONS OF SUPERCONDUCTOR

Coil No.	Wire Dia. mm	Filaments	Matrix	Crit. Current* A
A	1.0	830	OFHC	660
B	0.85	830	OFHC	470
C	0.75	830	OFHC	370

* the guaranteed critical current at 5T and 4.2K.

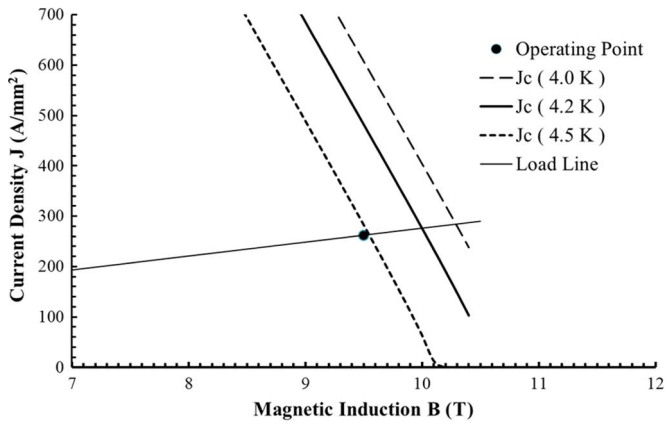

Fig. 3. Load line of coil A.

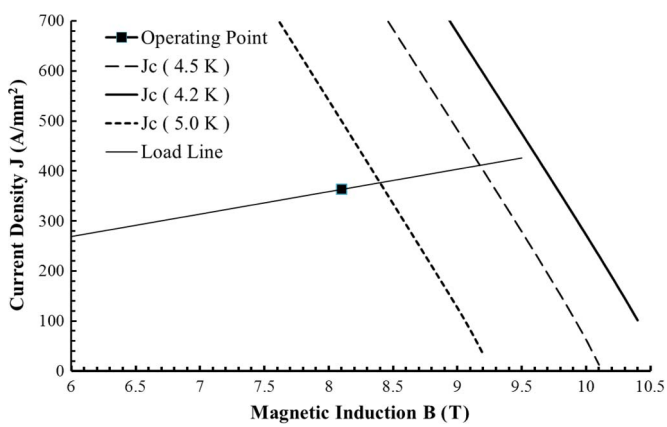

Fig. 4. Load line of coil B.

Conductors of the magnet coils are eventually determined as commercial products. Conductor specifications of the innermost three coils are listed in Table II.

The critical surface Jc(B,T) of NbTi conductor for coils A and B are fitted using the empirical formula [16] which accuracy are generally sufficient for design purposes. The current sharing temperature Tcs are evaluated as 4.53 K and 5.19 K, respectively, for coils A and B. Load lines of the innermost two coils A and B are depictured in Figs. 3 and 4. Operation margins of coils A and B are estimated as 5% and 16%, respectively, based on their individual critical current densities at 4.2 K.

C. Structures of the Superconducting Magnet

As discussed from the previous paragraph, the 9.2-T superconducting magnet is composed of totally ten NbTi coils and has a very stringent operation margin for coil A. This implies that magnet should be kept in a very uniform low temperature condition below the smallest current sharing temperature of 4.53 K by a GM cryocooler. Magnet structures should be carefully designed to guarantee temperatures of every coil lower than its current sharing temperature Tcs. Fig. 5 shows the magnet structure which integrated all the NbTi coils.

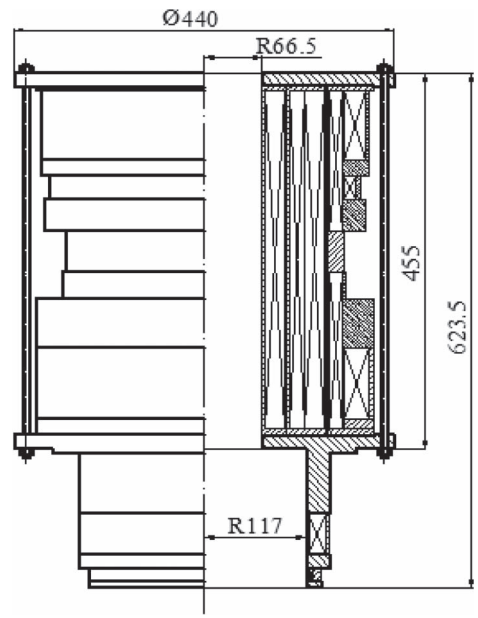

Fig. 5. Magnet structure.

The magnet structure is characterized as integration by four separate solenoids. Each solenoid is wound independently onto an aluminum alloy former. The integration is fulfilled by clasping all the solenoids with supporting bars. Soft metal with good ductility or cryogenic glue with high thermal conductivity can be applied between the mechanical contact surfaces to reduce thermal resistance.

This structure is in favor of axial thermal conduction to improve the temperature uniformity in coil windings. Actually, the magnet is a multi-material composite, including aluminum alloy, copper, NbTi and epoxy resin. The temperature uniformity of the coil depends on the parallel thermal conductivity of two paths. One is the coil former and the other is coil winding. The effective thermal conductivity λ_{eff} of coil windings can be evaluated utilizing the analogy method to electric circuit [17]. For the innermost coil A, the λ_{eff} is computed as 0.2 W/m $\cdot$ K in the transverse direction, which is much smaller than the thermal conductivity of its former materials at 4.2 K. So, the temperature uniformity is dominated by high conductive coil former which is subject to radiation loads from thermal shield. It is further estimated that an axial temperature drop of 0.3 K requires a radiation power density of 0.36 W/m^2 on the inner surface of coil A, which is not a problem for a carefully designed cryogenic system.

III. CRYOGENICS OF THE MAGNET SYSTEM

Cryogenics of the magnet system is very important to keep the coils in a temperature as low as possible. The conduction cooling scheme is adopted as the magnet cooling and low temperature holding. Cryogenics of the magnet system is schematically drawn in Fig. 6.

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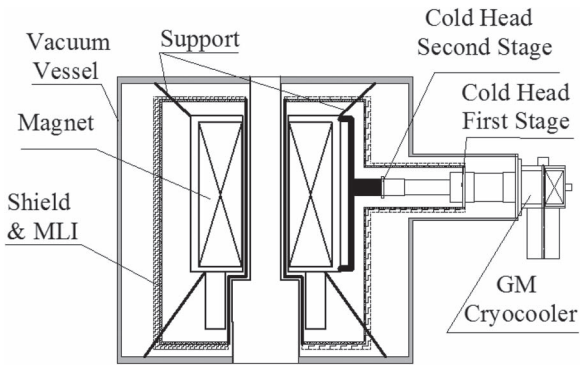

Fig. 6. Cryogenics of the magnet system.

TABLE III
THERMAL LOADS OF THE CRYOGENIC SYSTEM

Cold Head	Sources	Thermal Load (W)	Total (W)
First Stage	Radiation*	25.11	34.94
	Mechanical Support	0.38	
	Current Lead	9.45	
Second Stage	Radiation*	0.16	0.36
	Mechanical Support	0.02	
	Current Lead	0.18	

*the radiation power is estimated at the condition of shield temperature 45 K with 20 layers MLI.

The GM cryocooler is arranged horizontally with its driving motor at a location where the stray field is lower than 500 Gauss. Cooling powers of the GM cryocooler are commercially specified as 35 W at 50 K, 1.5 W at 4.2 K for the first and second cold heads, respectively. Thermal loads of the cryogenic system are calculated and listed in Table III.

It shows that the thermal loads of the cryogenic system are acceptable for the final engineering construction. It is possible to cool the magnet down to a temperature lower than 4.2 K, which provides an additional margin for the magnet operation.

IV. CONCLUSION

The 9.2-T NbTi superconducting magnet is designed and will be fabricated for high-frequency gyrotron experiments. The magnetic field and distribution can be used to accommodate a variety of gyrotron experiments. Magnetic analysis showed that the operation margin is close to the critical performance of commercial NbTi superconductors, which requires a carefully design and integration of the cryogenic system.

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